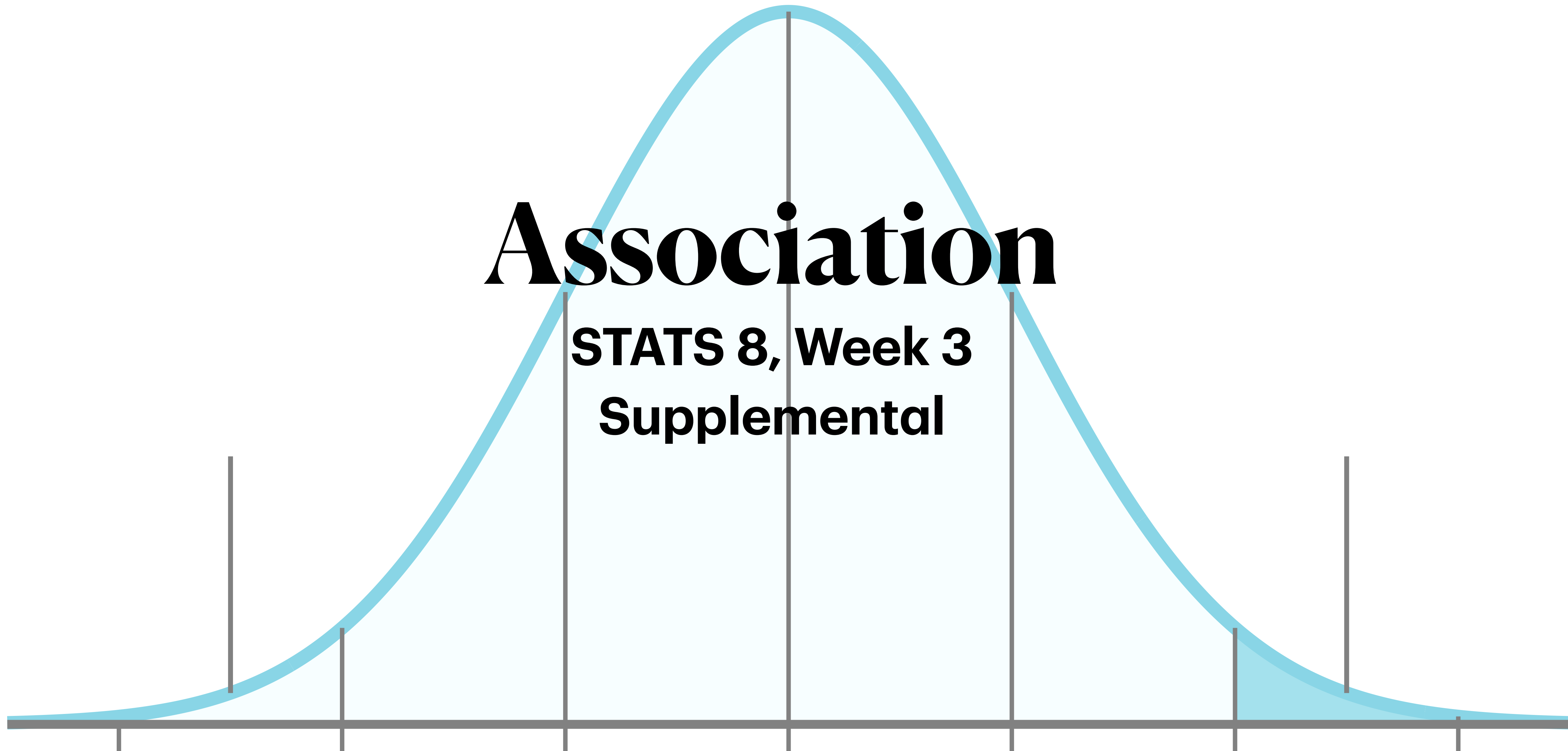
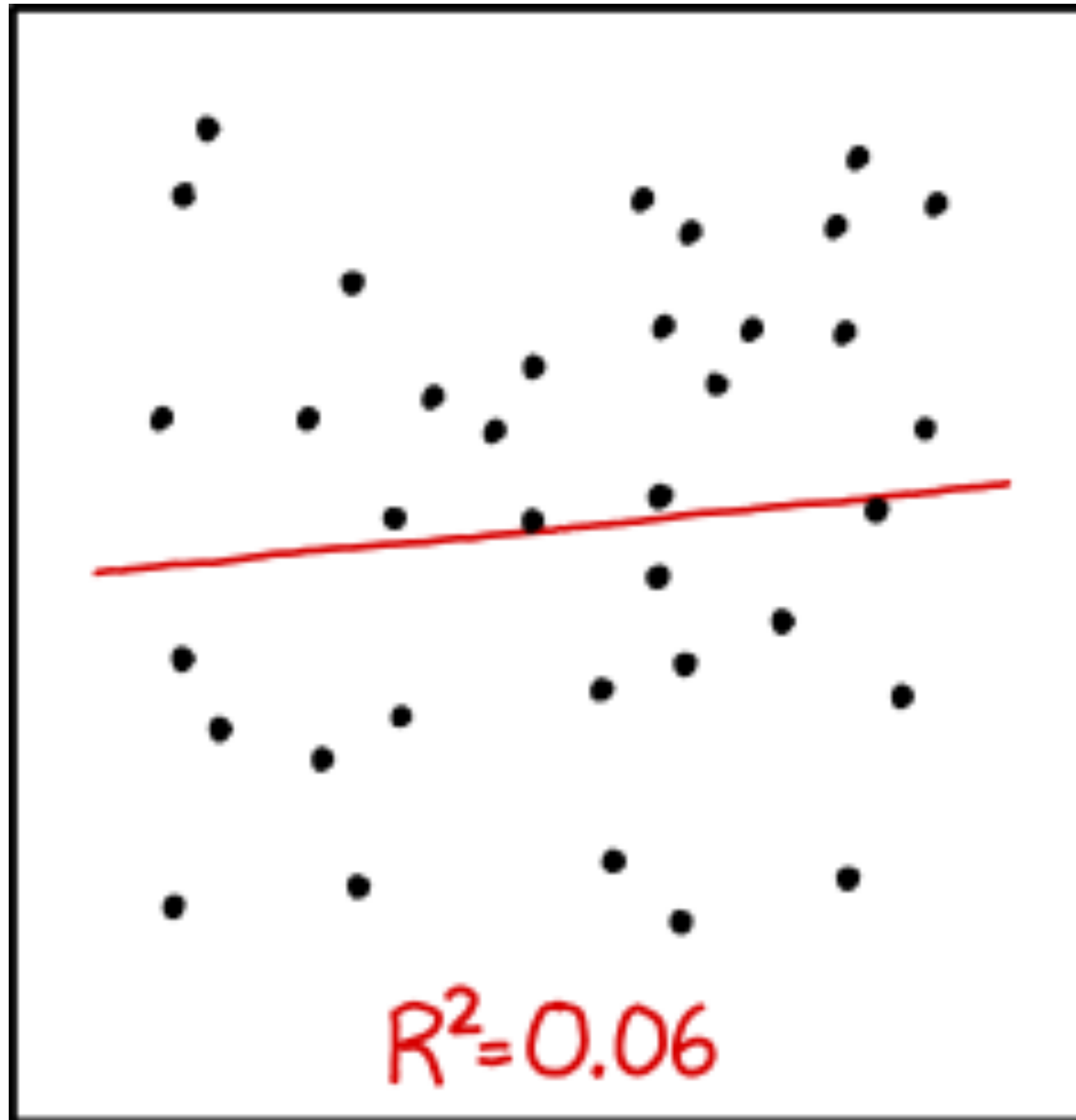


Association

**STATS 8, Week 3
Supplemental**





- PollEv.com/stats8
- What percentage of the variation in the response variable can be explained by the explanatory variable?

I DON'T TRUST LINEAR REGRESSIONS WHEN IT'S HARDER
TO GUESS THE DIRECTION OF THE CORRELATION FROM THE
SCATTER PLOT THAN TO FIND NEW CONSTELLATIONS ON IT.

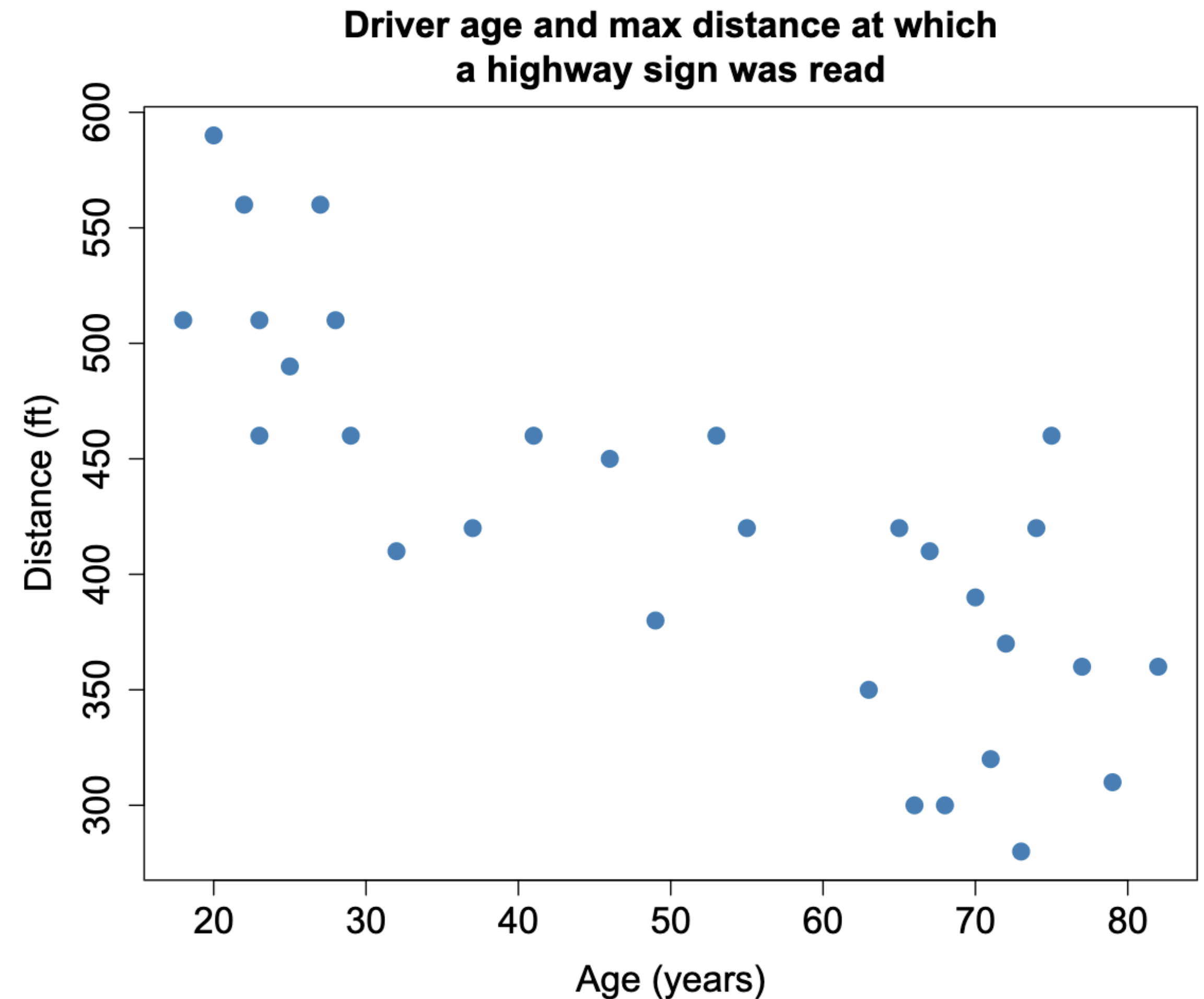
(xkcd.com)

Example: Correlation

PollEv.com/stats8

- The correlation coefficient r is...

- A. Positive
- B. Approximately 0
- C. Negative

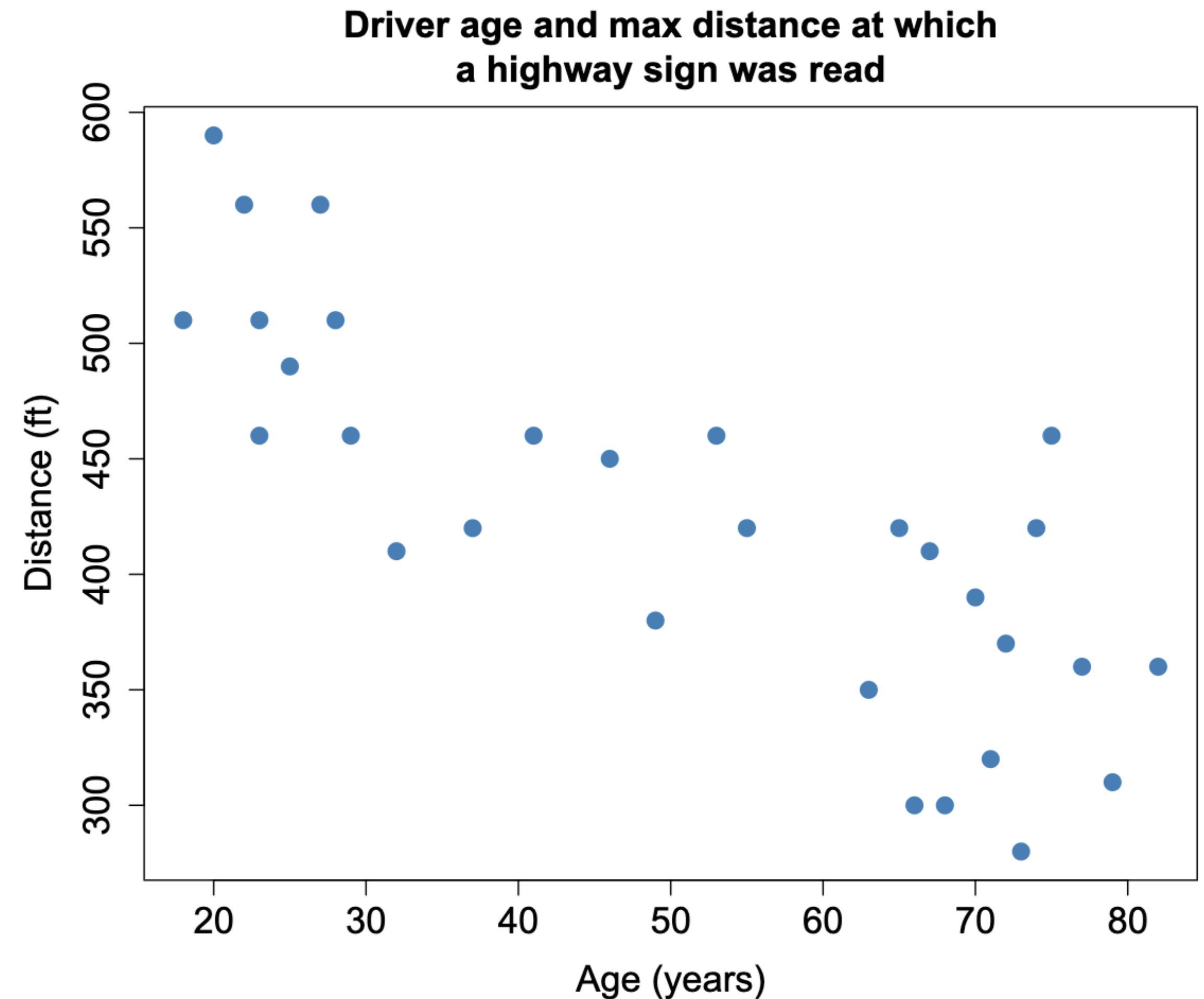


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PollEv.com/stats8

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Example: Correlation

PollEv.com/stats8

- The correlation coefficient r is -0.80.
- What percentage of the variability in distance at which a person can read a highway sign is explained by age?



Example: Correlation

PollEv.com/stats8

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$$(-0.8)^2 = 0.64 = 64 \%$$



Example: Correlation

PollEv.com/stats8

- The correlation coefficient r is -0.80.
- How would the correlation coefficient change if we measured distance in yards (3 feet) instead of feet?
 - A. It would get closer to -1
 - B. It would get closer to 0
 - C. It would not change



Example: Correlation

PollEv.com/stats8

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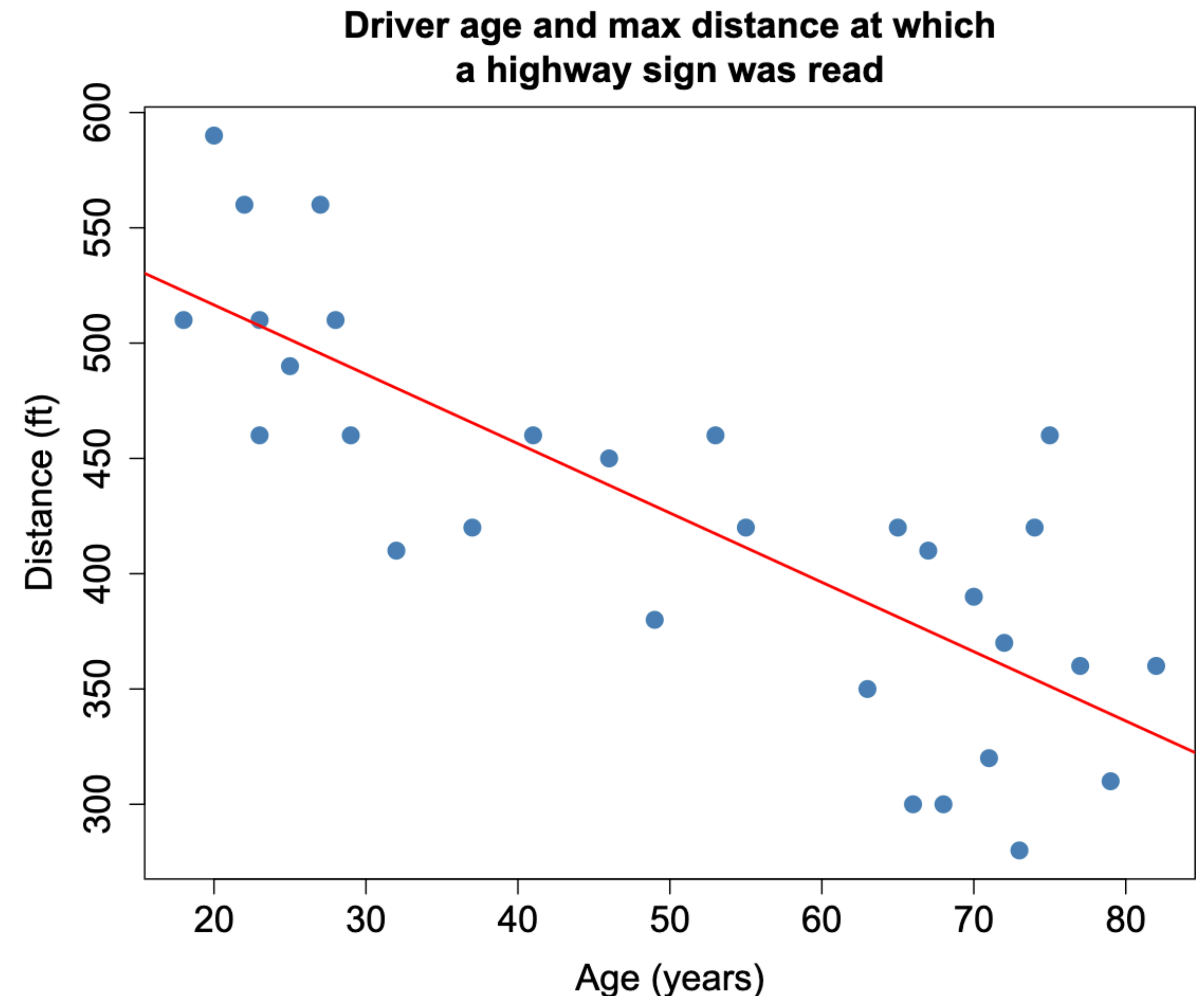
Example: Regression

PollEv.com/stats8

- The regression line describing this relationship is given by

$$(\text{dist}) = 577 - 3 \times (\text{age})$$

- What is the interpretation of the intercept in this example?



Example: Regression

PollEv.com/stats8

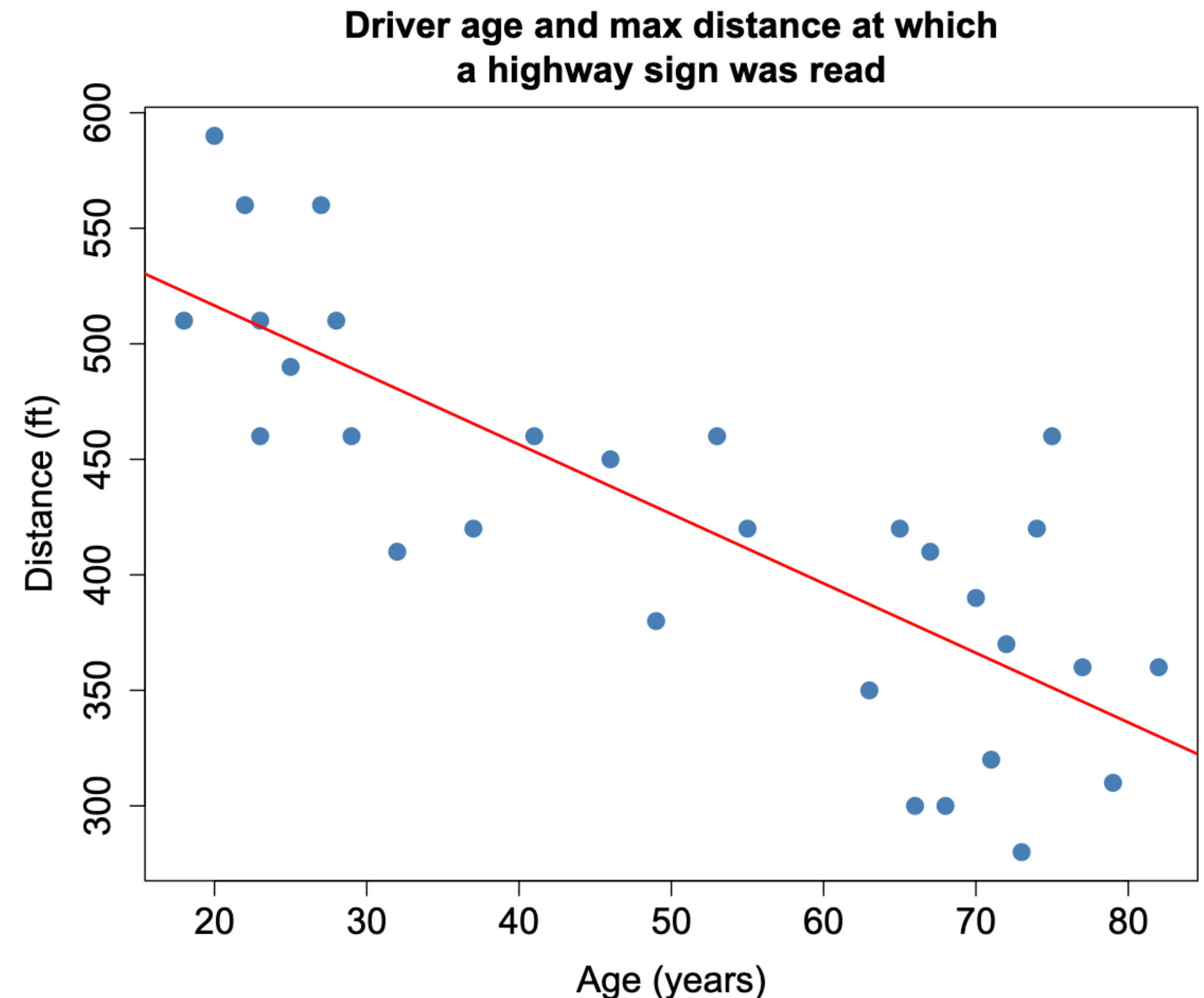
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- What is the interpretation of the intercept in this example?

The average max distance at which people 0 years old can read a highway sign is 577 feet.

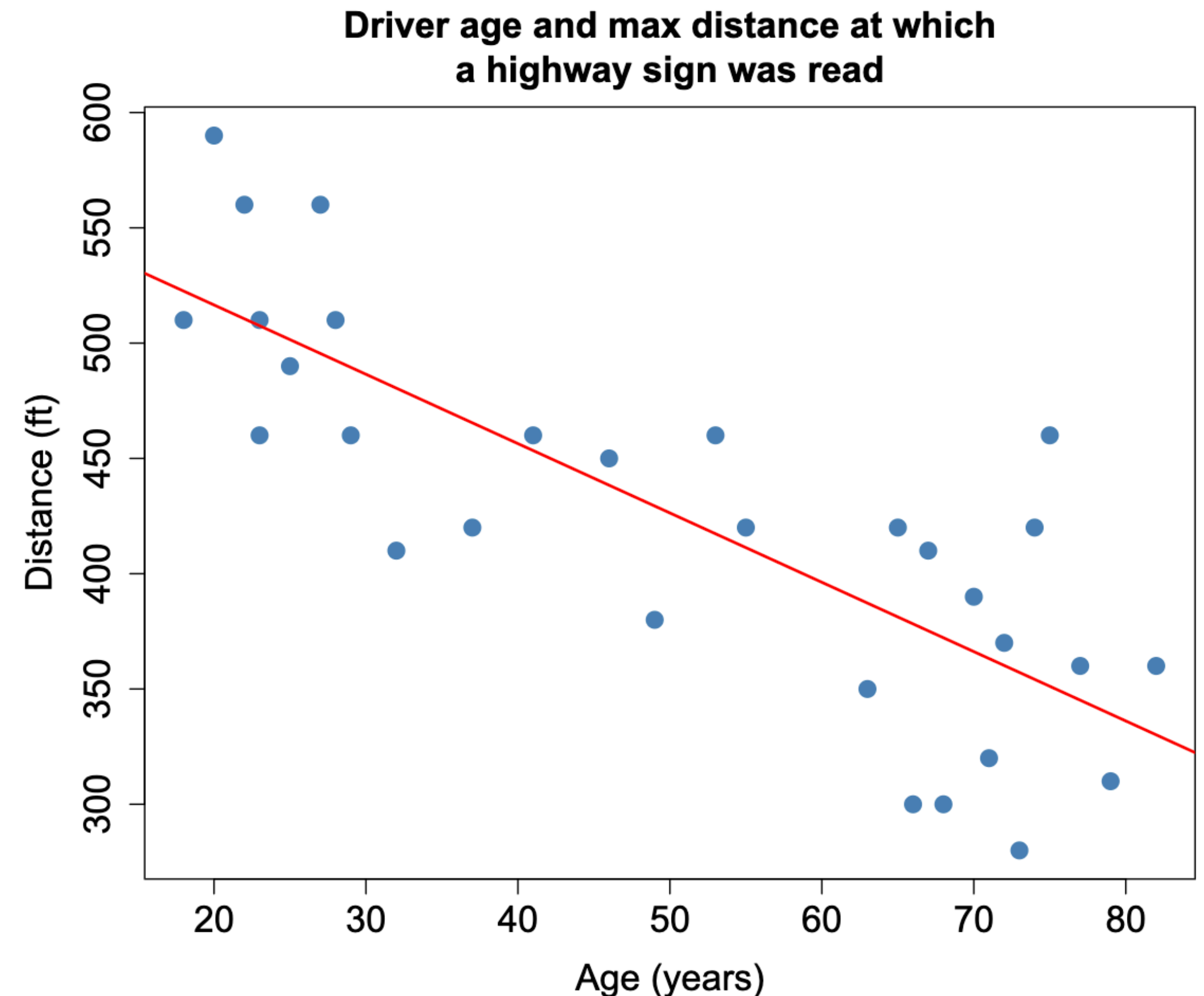
But... this is not practically interpretable, because it's extrapolation & doesn't make sense in context.



Example: Regression

PollEv.com/stats8

- The regression line describing this relationship is given by
$$(\text{dist}) = 577 - 3 \times (\text{age})$$
- What is the interpretation of the slope in this example?



Example: Regression

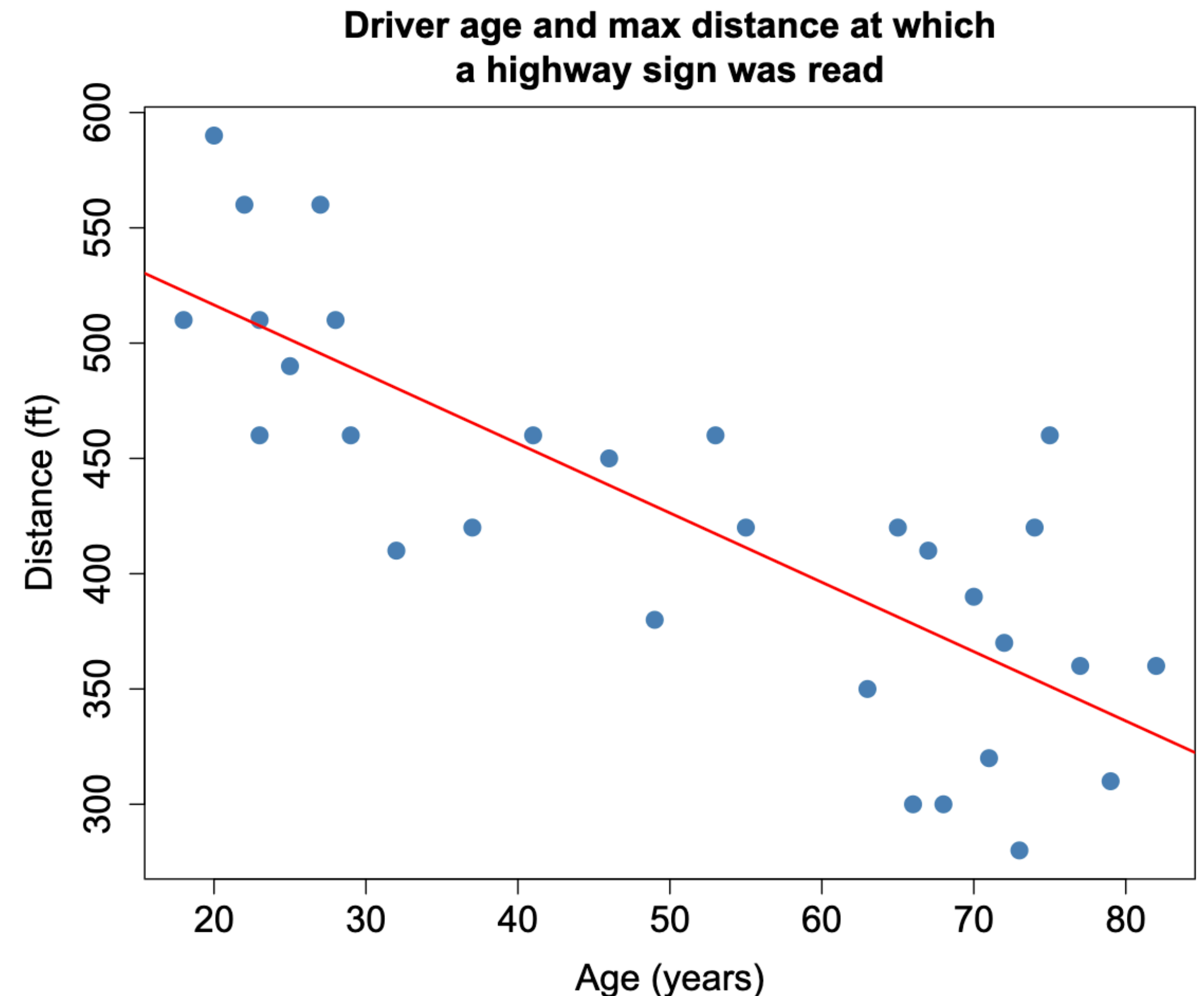
PollEv.com/stats8

- The regression line describing this relationship is given by

$$(\text{dist}) = 577 - 3 \times (\text{age})$$

- What is the interpretation of the slope in this example?

Comparing two groups of people 1 year apart in age, the older group has a 3 foot lower average max distance at which they can read a highway sign.



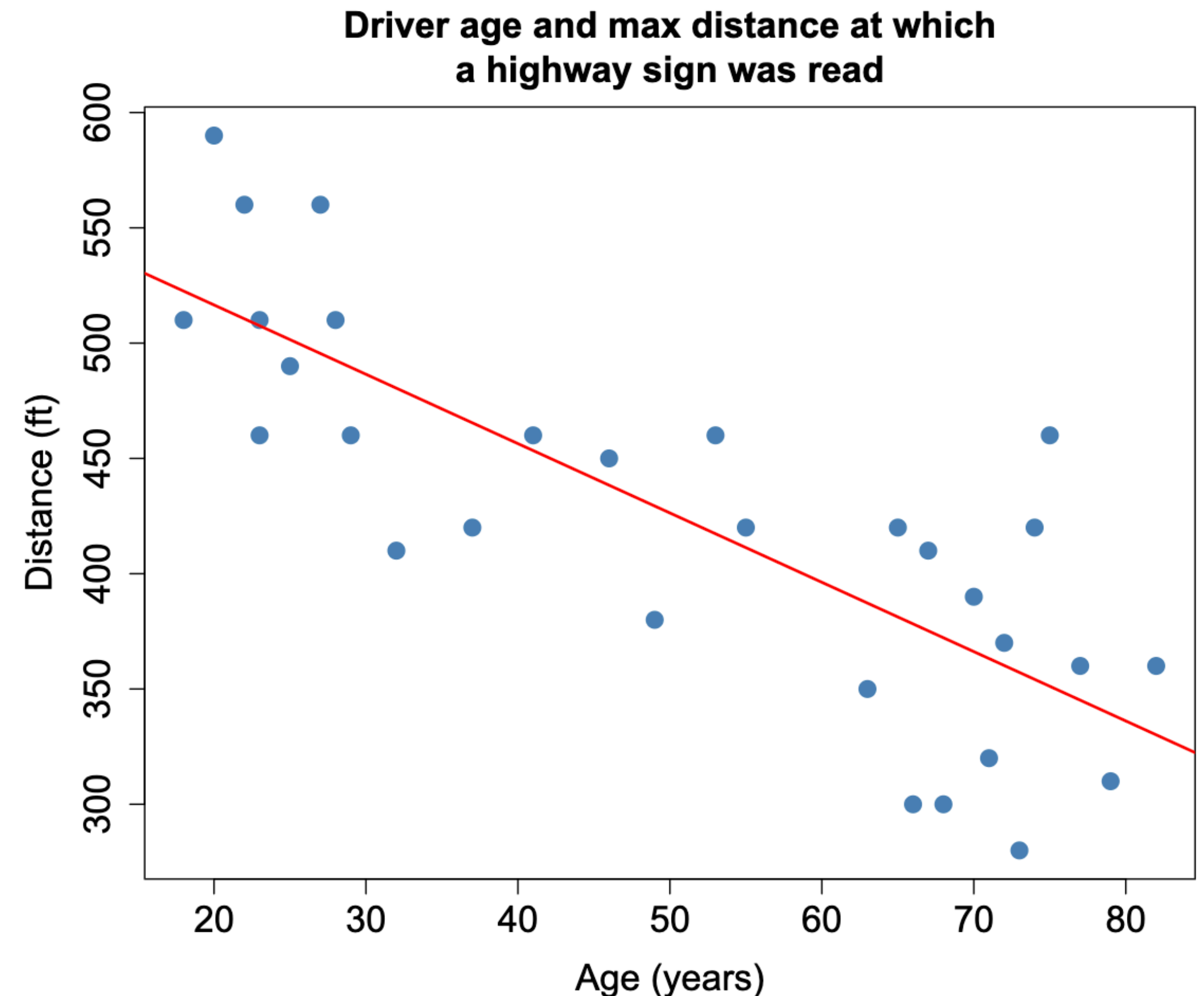
Example: Regression

PollEv.com/stats8

- Recall that a residual is given by

$$e = y - \hat{y}$$

- Which points have negative (less than 0) residuals?
 - The points above the line
 - The points below the line
 - None, residuals can't be negative



Example: Regression

PollEv.com/stats8

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